

Method 1 (Baseline): YOLOv11 + SORT

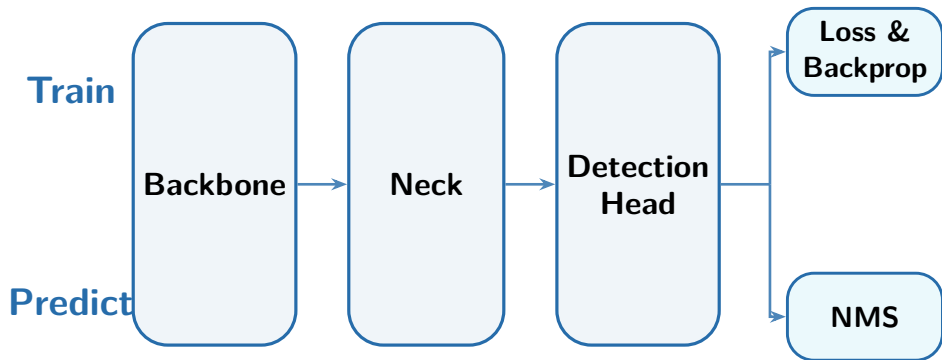
Computer Vision — UTEC

PC4: Detection and Tracking

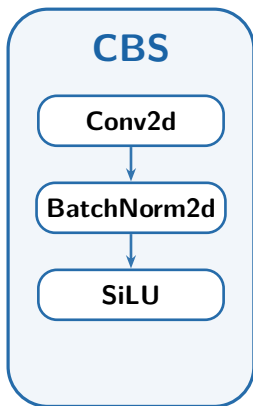
Rensso Victor Hugo Mora Colque

2026-I

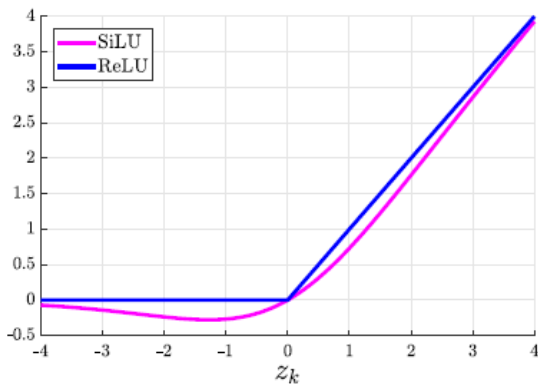
YOLOv11 Architecture

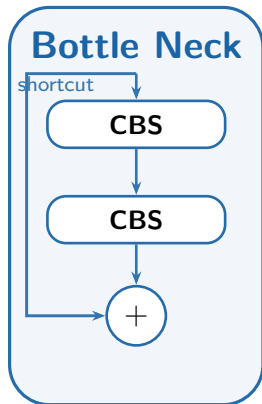


YOLOv11: Conv + BatchNorm + SiLU Block



$$\text{SiLU}(x) = x \cdot \frac{1}{1 + e^{-x}}$$





Bottleneck idea

- ▶ Two CBS layers transform the feature map with low extra cost.
- ▶ The shortcut adds the input back to the output:

$$\mathbf{y} = F(\mathbf{x}) + \mathbf{x}$$

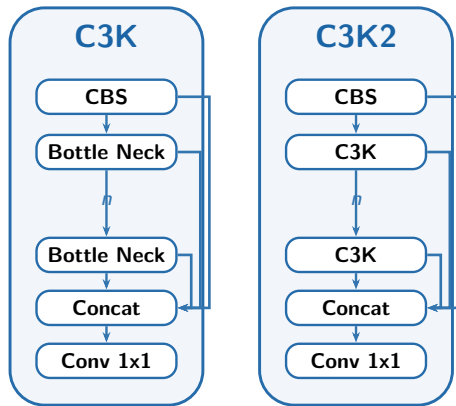
- ▶ Same residual principle as a ResNet block: preserve useful input information and improve gradient flow.
- ▶ CBS means **Conv + BatchNorm + SiLU**; convolution extracts features, BatchNorm stabilizes training, and SiLU adds nonlinearity.

C3K block

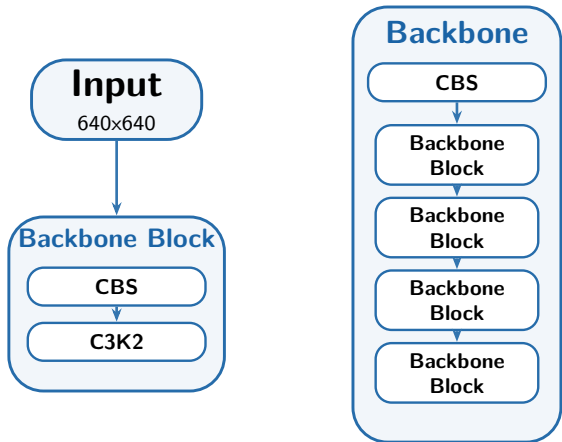
- ▶ Repeats bottleneck blocks n times to deepen feature extraction.
- ▶ Final concat keeps both transformed and preserved features.

C3K2 improvement

- ▶ Replaces each repeated bottleneck with a smaller C3K unit.
- ▶ Keeps multi-stage feature reuse while reducing cost.



YOLOv11: Backbone Step

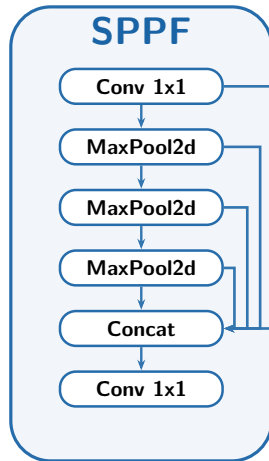


Backbone role

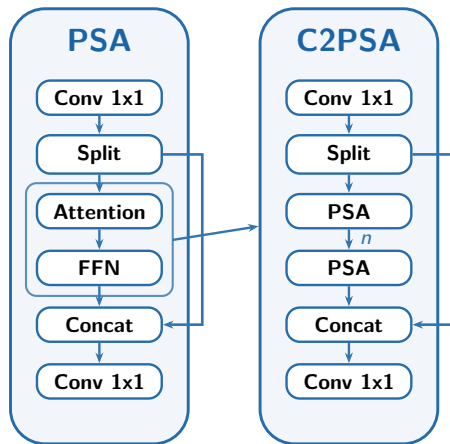
- ▶ Receives the input image and extracts visual features at increasing depth.
- ▶ Early CBS layer detects simple patterns such as edges, colors, and small textures.
- ▶ Each stage reduces spatial resolution and increases semantic information for detection.
- ▶ Output feature maps feed the neck

YOLOv11: Spatial Pyramid Pooling Fast

- ▶ It expands the receptive field before features are passed to the neck.
- ▶ Repeated max-pooling captures context at several spatial scales.
- ▶ Concat merges local and pooled features so the neck can fuse richer multi-scale information.
- ▶ It is efficient because sequential pooling reuses intermediate outputs instead of computing separate large kernels.



YOLOv11: Cross Stage Partial Self-Attention

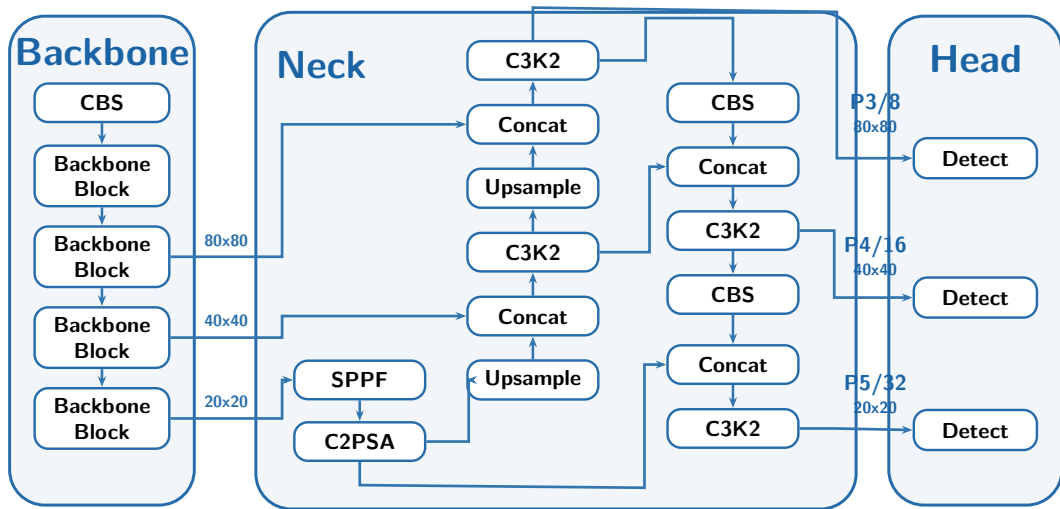


Attention core

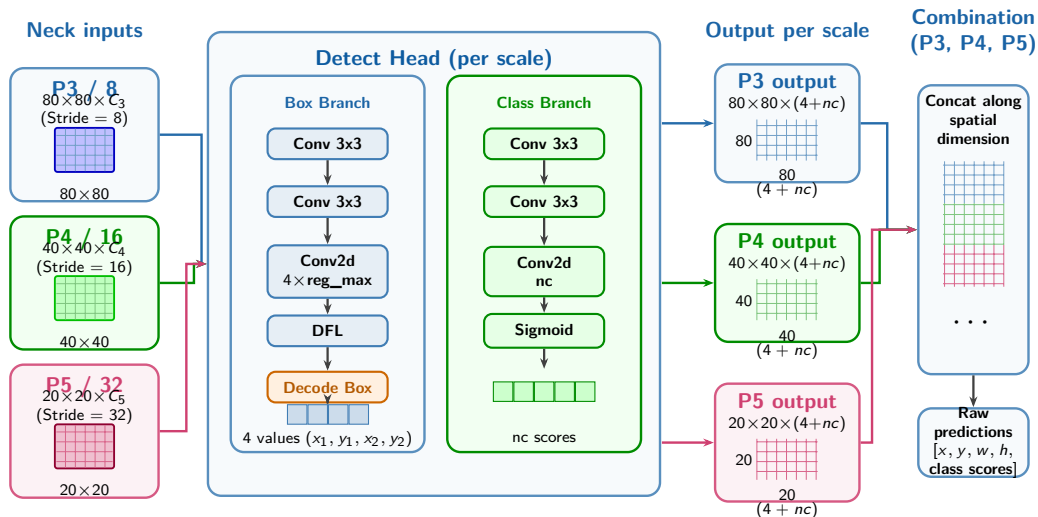
$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- ▶ QK^T scores how strongly each position should attend to other positions.
- ▶ $\sqrt{d_k}$ stabilizes score scale before softmax.
- ▶ PSA adds global spatial context, then FFN refines channel features.
- ▶ C2PSA first splits channels, applies PSA on one path, then concatenates and compresses with Conv 1×1 .

YOLOv11: Neck Step



YOLOv11: Detection Head Step



YOLOv11: Distribution Focal Loss and Decode Box

DFL (Distribution Focal Loss)

Example: left side

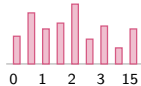
reg_max = 16

Bins (discrete distances)



Softmax

Probabilities



DFL expectation

$$d_{\text{left}} = \sum_{i=0}^{16-1} P_i \cdot i$$

DFL output

(4 distances)



Applied independently to left, top, right, bottom

Decode Box (per cell)

Inputs

Anchor
(x_a, y_a)



Distances
(l, t, r, b)

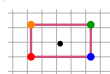


Formulas

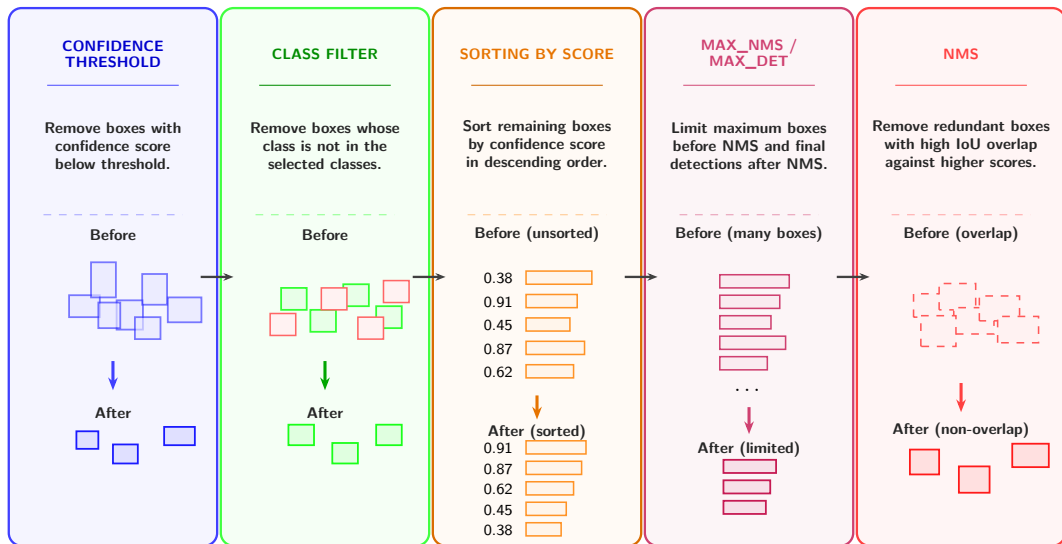
$$\begin{aligned}x_1 &= x_a - l \\y_1 &= y_a - t \\x_2 &= x_a + r \\y_2 &= y_a + b\end{aligned}$$

Output

(x_1, y_1, x_2, y_2)



YOLOv11: Postprocessing and NMS

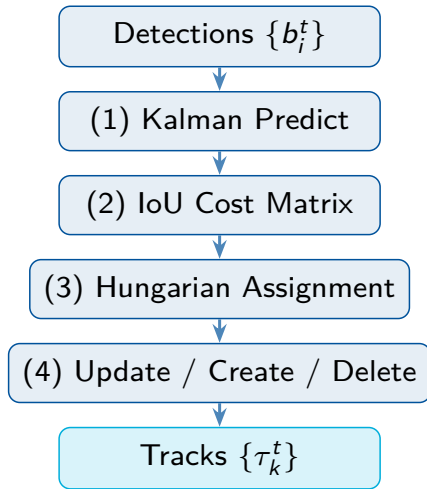


Bewley et al., ICIP 2016 [2]:

- ▶ Tracking-by-detection paradigm: tracker receives bboxes from the detector, does not re-detect
- ▶ **No visual appearance features** — purely geometric
- ▶ Two classical components:
 - ▷ **Kalman Filter**: motion prediction
 - ▷ **Hungarian Algorithm**: optimal assignment
- ▶ Achieves > 260 FPS on tracking alone

Why SORT as baseline?

Has limitations in long occlusion, appearance ambiguity, tiny objects.



SORT: Kalman Filter for Motion Prediction

State vector (7-dimensional, one Kalman filter per track):

$$\mathbf{x} = [c_x, c_y, s, r, \dot{c}_x, \dot{c}_y, \dot{s}]^\top$$

where c_x, c_y : bbox centre · $s = w \cdot h$: area · $r = w/h$: aspect ratio (assumed constant)
· $(\dot{\cdot})$: velocity

Prediction step (before observing detections):

$$\mathbf{x}_t^- = F \mathbf{x}_{t-1}, \quad P_t^- = F P_{t-1} F^\top + Q$$

F : constant-velocity transition matrix

P : state covariance Q : process noise

Update step (after associating a detection $\mathbf{z} = [c_x, c_y, s, r]^\top$):

$$K = P_t^- H^\top (H P_t^- H^\top + R)^{-1}$$

$$\mathbf{x}_t = \mathbf{x}_t^- + K(\mathbf{z} - H \mathbf{x}_t^-), \quad P_t = (I - KH) P_t^-$$

H : observation matrix R :

measurement noise K : Kalman gain
(weights prediction vs. measurement)

SORT: Hungarian Assignment and Track Lifecycle

IoU-based cost matrix:

$$C_{ij} = 1 - \text{IoU}(\hat{\tau}_i, b_j^t)$$

where $\hat{\tau}_i$: Kalman-predicted bbox of track i , b_j^t : detection j at time t

Hungarian algorithm minimises total cost:

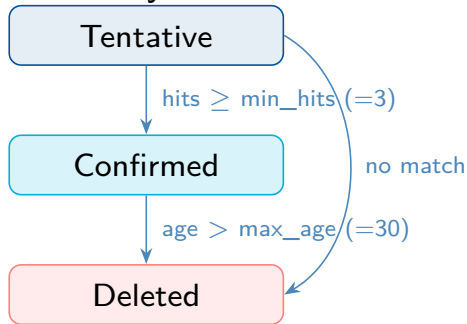
$$\sigma^* = \arg \min_{\sigma} \sum_i C_{i, \sigma(i)}$$

Match rejected if $\text{IoU} < \tau_{\text{IoU}} = 0.3$

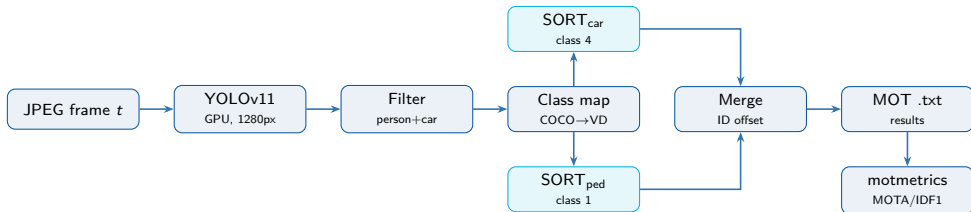
Separate trackers per class

Independent SORT instances for pedestrians and cars prevent cross-class identity confusion.

Track lifecycle:



Full Pipeline: YOLOv11 + SORT



Key design choice

Separate SORT trackers per class. Car IDs offset by +100 000 to prevent ID collision across classes in the output file.

Output format (MOT standard)

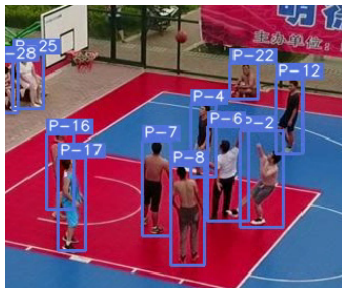
frame, id, x, y, w, h, conf, class, vis
One line per detection per frame. Sorted by (frame, id).

Qualitative Demo — Success Case (Track Continuity)

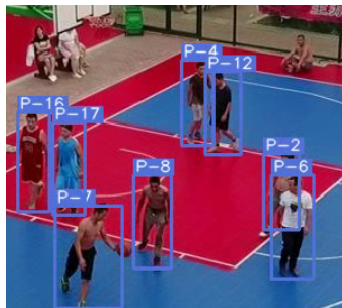
Frame t



Frame $t + 15$



Frame $t + 30$

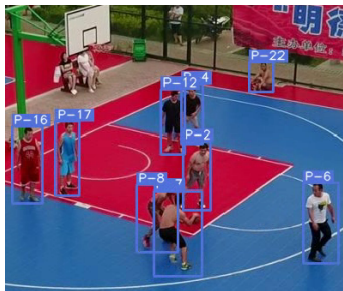


Look!

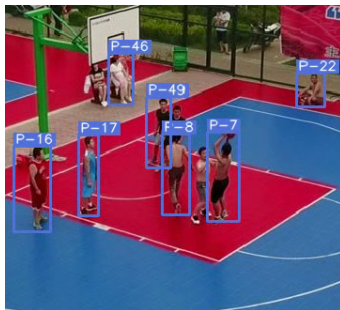
Same coloured bounding boxes and numeric IDs across the three frames, demonstrating that SORT maintains identity through motion.

Qualitative Demo — Failure Case (ID Switch)

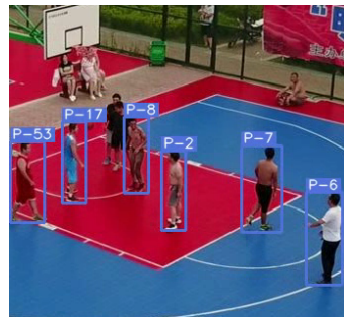
Before occlusion



During occlusion



After occlusion



ID Switch

When an object is occluded for $> \text{max_age}$ frames, SORT deletes the track. On reappearance, no Kalman prediction is available $\Rightarrow$ a new ID is assigned.

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-  T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common Objects in Context," in *Proc. European Conf. Computer Vision (ECCV)*, Zurich, Switzerland, Sep. 2014, pp. 740–755. DOI: 10.1007/978-3-319-10602-1_48.
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-  H. W. Kuhn, "The Hungarian Method for the Assignment Problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955. DOI: 10.1002/nav.3800020109.
-  N. Wojke, A. Bewley, and D. Paulus, "Simple Online and Realtime Tracking with a Deep Association Metric," in *Proc. IEEE Int. Conf. Image Processing (ICIP)*, Beijing, China, Sep. 2017, pp. 3645–3649. DOI: 10.1109/ICIP.2017.8296962.
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-  O. F. Cengiz, F. Akyon, E. Altinuc, and A. Tekin, "Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection," in *Proc. IEEE Int. Conf. Image Processing (ICIP)*, Bordeaux, France, Oct. 2022, pp. 966–970. DOI: 10.1109/ICIP46576.2022.9897990.

Thank you

Questions?

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Assignment IV: Detection and Tracking — Method 1 Baseline